Theorem Book - Analysis

Ryan Lam

May 10, 2021

1 Introduction

This is a place to record important theorem that I should remember by heart.

2 Peano Axioms, The Natural Numbers

The axiom ensures the existence of a set called **natural numbers,** denoted by $\mathbb{N}$.

Axiom 2.1. 0 is a natural number.

Axiom 2.2. If n is a natural number, then $n{+}{+}$ is also a natural number.

Axiom 2.3. 0 is not the successor of any natural number. That is, for all natural numbers n, m,

$$n = m++ \Rightarrow n \neq 0$$

Axiom 2.4. For all natural numbers m, n, $m{+}{+} \neq n{+}{+} \Leftrightarrow m \neq n$

Axiom 2.5 (Induction)**.** If there is a subset S of of $\mathbb{N}$, and the subsets have the following properties:

- $0 \in S$
- For a natural number $n, n \in S \Rightarrow n++ \in S$

Then the set is $\mathbb{N}$ itself.

Definition 2.6. The **addition** on two natural number m, n is defined as:

$$0 + m = m \qquad \text{and} \qquad (n++) + m = (n + m)++$$

Definition 2.7. For all natural numbers m, n, write $n \geqslant m$, and say n is **larger than or equal to** m, when $n = m + a$ for some natural number a. When a is not zero, we say n is **greater than** m and write $n > m$.

Theorem 2.8. $\geqslant$ forms a **partial order** on the natural number $\mathbb{N}$. That is, for all $a, b, c \in \mathbb{N}$:

- (Reflexive) $a \leqslant a$
- (Transitive) If $a \leqslant b, b \leqslant c$, then $a \leqslant c$.
- (Anti-symmetric) If $a \leqslant b, b \leqslant a$, then $a = b$.

The mnemonic **RoTA** can helps to remember. It also preserves addition. That is, $a \geqslant b \Rightarrow a + c \geqslant b + c$ for all natural number c.

Theorem 2.9 (Trichotomy)**.** Exactly one of the three situation can happens for natural number a, b: $a = b, a > b$ and $a < b$.

Definition 2.10. Multiplication of two numbers, m, n, are recursively defined as:

$$0 \times m = 0 \qquad \text{and} \qquad (n++) \times m = (n \times m) + m$$

Theorem 2.11. The natural numbers are **associative, commutative, closed** under addition and multiplication, they also follows the distributive law: $a(b+c) = ab + ac$.

Theorem 2.12 (Division Algorithm)**.** Given a non-zero natural number q, then for all natural number m we can write:

$$m = bq + r \qquad \text{with} \qquad 0 \leqslant r < q$$

3 Axiom on Real Numbers

Definition 3.1. A **group** is a set G attached with a binary operation $*$ with the following property holds for all $a, b, c \in G$:

$$a, b \in G \Rightarrow a * b \in G \qquad \text{(Closed binary Operation)}$$
$$(a * b) * c = a * (b * c) \qquad \text{(Associativity)}$$

There exists an (unique) element $e \in G$ such that:

$$e * a = a * e = e \qquad \text{(Identity)}$$

For all $a \in G$ there exists an element, denoted by a^{-1}, such that:

$$a * a^{-1} = a^{-1} * a = e \qquad \text{(Inverse)}$$

Definition 3.2. A group $\langle G, * \rangle$ is **Abelian** if and only if for every element $a, b, \in G$, we have $a*b = b*a$.

Definition 3.3. The set of rational number $\mathbb{Q}$ can be represented by 4 natural number $(a, b, c, d) \in \mathbb{N}^4$, denoted by $(a - -b)//(c - -d)$, with $c \neq d$.

There are not only one choice of (a, b, c, d) such the the value of $q = (a, b, c, d)$ is the same. Note that $(a - -b)$ correspond to subtraction and $//$ correspond to division in the usual sense

Theorem 3.4. Assume that addition and multiplication of $\mathbb{N}$ was defined as above, then if we define the addition of two rational numbers to be $+^{\mathbb{Q}}$:

$$[(a--b)//(c--d)]+^{\mathbb{Q}}[(e--f)//(g--h)] = (ag+bh+ec+fd--ah+bg+ed+fc)//(cg+dh--ch+dg)$$

And multiplication $\times^{\mathbb{Q}}$ defined as:

$$[(a - -b)//(c - -d)] \times^{\mathbb{Q}} [(e - -f)//(g - -h)] = (ae + bf - -af + be)//(cg + dh - -ch + dg)$$

Then those operation are well-defined.

Definition 3.5. A **field** is a set F equipped with two binary operation $\langle F, +^F, *^F \rangle$, where:

- $\langle F, +^F \rangle$ is an abelian group, we denoted the identity to be 0^F.
- $\langle F \setminus \{0^F\}, *^F \rangle$ is an abelian group, we denote the identity to be 1^F.
- The distributive law: $a *^F (b +^F c) = a *^F b +^F a *^F c$ holds.

Definition 3.6. An **ordered field** is a where field there exists a relation $<$ and the four axiom of ordering holds for all x, y, z in the field,:

- (Trichotomy) Exactly one of the three is true: $x > 0, x = 0, x < 0$
- (Compatibility with addition) $x < y \Rightarrow x + z < x + y$
- (Compatibility with multiplication) $(x < y) \wedge (z > 0) \Rightarrow xz < yz$
- (Transitivity) $(x < y) \wedge (y < z) \Rightarrow x < z$

Theorem 3.7. $\langle \mathbb{Q}, +^{\mathbb{Q}}, \times^{\mathbb{Q}} \rangle$ with the usual ordering $>$ is an ordered field.

4 Absolute value, Supremum and Infimum

From this point onward, it is a quick conclusion for the course MATH10011 - Analysis.

Definition 4.1. A set A is **bounded above** if for all $a \in A$, there exists some $k \in \mathbb{Q}$ such that $k > a$.
A set A is **bounded below** if for all $a \in A$, there exists some $k \in \mathbb{Q}$ such that $a > k$.
A set which is bounded above and below is **bounded**.
A set which has no upper bound and lower bounded are called **unbounded above** and **unbounded below** respectively.

Definition 4.2. A **least upper bound** for a set A is a number b such that b is the upper bound of A, and for all upper bound k we have $b \leqslant k$.
A **greatest lower bound** for a set A is a number b such that b is the lower bound of A, and for all upper bound k we have $b \geqslant k$.

Axiom 4.3. The set $\mathbb{R}$ is $\mathbb{Q}$ with the **least upper bound property**: every set which is bounded above have a least upper bound.

Using the fact that $\mathbb{N}$ is unbounded, we have

Theorem 4.4 (Archimedean Property)**.** For every $\epsilon > 0$, there exists $n \in \mathbb{N}$ such that $\frac{1}{n} < \epsilon$

Definition 4.5. The **absolute value** of a real number x, denoted by $|x|$, is

$$|x| = \begin{cases} x & \text{when } x > 0 \\ -x & \text{when } x \leqslant 0 \end{cases}$$

Theorem 4.6 (Triangle inequality)**.** Probably the most important in this course, for all $x, y, z \in \mathbb{R}$:

- $|x + y| \leqslant |x| + |y|$
- $|x - y| \leqslant |x - z| + |z - y|$
- $||x| - |y|| \leqslant |x - y|$

Definition 4.7. The **supremum** of a set A always exists, denoted by $\sup A$, it is:

- ∞ if the set is unbounded above.
- $-\infty$ if $A = \emptyset$
- The least upper bound of A otherwise.

Similarly, **infimum** of a set A always exists, denoted by $\inf A$, it is:

- $-\infty$ if the set is unbounded below.
- ∞ if $A = \emptyset$
- The greatest lower bound of A otherwise.

To find the supremum of a set, first conjecture the supremum, if it is ∞ then prove it is unbounded, else, show there is an upper bound, then you have two ways to proceed:

1. Show that if $b < \sup A$ then there exists $a \in A$ such that $a > b$, or
2. For all $\epsilon > 0$ there exists $a \in A$ such that $\sup A - \epsilon < a$

Which the second way means that "$\sup A$ move left a bit will touch inside A". For infimum, we reproduce similar procedure. Archimedean property while manipulating the inequality.

5 Sequences

Definition 5.1. A **sequences** is a function from natural numbers to real number $f(n) = a_n$, or we might write

$$(a_n)_{n\in\mathbb{N}} = (a_1, a_2, a_3...)$$

The above definition will coincide with later discussion of continuity.

Definition 5.2. A sequences is **bounded** if there exists $M \in \mathbb{R}$ such that $a_i \leqslant M$ for all M. else it is **unbounded**.

Definition 5.3. A sequence is **convergent to limit** L, and we write $\lim_{n\to\infty} a_n = L$ if for every $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$n \geqslant N \Rightarrow |a_n - L| \leqslant \epsilon$$

Intuition: for the sequence past a 'moment' N, we can get as close (ϵ) as we want.

Theorem 5.4. Every convergent sequence is bounded.

Finding a limit can also be done my manipulating the right hand side of the equation first (usually by triangle inequality), then write the answer backwards.

Theorem 5.5 (Limit Laws)**.** If for sequences $(a_n), (b_n)$ we have $\lim_{n\to\infty} a_n = L$, $\lim_{n\to\infty} b_n = M$, then

- $\lim_{n\to\infty}(a_n + b_n) = L + M$, if $L = M = \pm\infty$ then $\lim_{n\to\infty}(a_n + b_n) = \pm\infty$
- $\lim_{n\to\infty}(a_n)(b_n) = LM$, if $L = M = \pm\infty$ then $\lim_{n\to\infty}(a_n b_n) = \pm\infty$ (Note that $-\infty \times +\infty, \infty \times 0$ the theorem is inconclusive)
- $\lim_{n\to\infty}\left(\frac{a_n}{b_n}\right) = \frac{L}{M}$ if $M \neq 0, b_n \neq 0$ for all $n \in \mathbb{N}$
- (Sandwich Rule) If $a_n \leqslant c_n \leqslant b_n$ for all $n \in \mathbb{N}$ and $L = M$, then $\lim_{n\to\infty} c_n = L$, most notably we have

  $$\lim_{n\to\infty} \frac{1}{n^k} = 0$$

  for all $k \geqslant 1$.

Definition 5.6. A sequence is said to **diverges to infinity** and we write $\lim_{n\to\infty} a_n = \infty$ if for all $k \in \mathbb{R}$ there exists $N \in \mathbb{N}$ such that

$$n \geqslant N \Rightarrow a_n > K$$

but if for every $K \in \mathbb{R}$ we have

$$n \geqslant N \Rightarrow a_n < K$$

then we say the sequences **diverges to negative infinity** and write $\lim_{n\to\infty} a_n = -\infty$.

Definition 5.7. A sequence a_m is **strictly monotone increasing** if for every $m, n \in \mathbb{N}$ with $m > n$ we have $a_m > a_n$, **strictly monotone decreasing** if we have $a_m < a_n$. In both cases they are called **strictly monotone** sequences. (monotone are $<$ replaced by $\leqslant$)

Theorem 5.8. Every monotone bounded sequence is convergent.

There is a very useful theorem handling recurring sequence, which is:

Theorem 5.9. For every constant $k \in \mathbb{N}$, $\lim_{n\to\infty} a_{n+k} = \lim_{n\to\infty} a_n$. (Works for infinity too)

So for example (5.9) given recurring sequence $a_{n+1} = \frac{5a_n + 2}{2a_n + 1}$. Given the sequence is increasing and bounded (prove by induction), then the limit $\lim_{n\to\infty} a_n = \alpha$ can be found by the equality $\alpha = \frac{5\alpha + 2}{2\alpha + 1}$. One can also prove the sequence $\lim_{n\to\infty} \lambda^n = 0$ when $|\lambda| < 1$ using similar strategy.

6 Subsequence, Cauchy

Definition 6.1. Let $(n_k)_{k\in\mathbb{N}}$ be a strictly increasing sequences, given a sequence $(a_n)_{n\in\mathbb{N}}$, the **subsequence** of is (a_{n_k}), for example, a_{2n} is such a subsequence.

We present a lot of confusing theorems regarding them. A rule of thumb is, if there is a weak condition on subsequence, then you can build a very weird sequence by inserting peculiar elements indefinitely.
The contrapositive of theorem might also be useful sometimes.

Theorem 6.2. A convergent sequence $\implies$ **all** subsequences are convergent. (Including 'convergence' to infinity)

Theorem 6.3. Unbounded Sequence $\implies$ **exists** subsequence (a_{n_k}) such that $\lim_{k\to\infty} |(a_{n_k})| = +\infty$

Theorem 6.4. $\lim_{n\to\infty} |(a_n)| = +\infty \Leftrightarrow$ **all** subsequence of (a_n) are **NOT** convergent.

Theorem 6.5 (Bolzanno-Weierstrass)**.** Bounded Sequence $\Rightarrow$ **exists** convergent subsequence

We then introduce a definition which makes proving limits easier

Definition 6.6. A sequence is **cauchy** if for every $\epsilon > 0$, there exists $N \in \mathbb{N}$, such that

$$m, n \geqslant N \implies |a_m - a_n| \leqslant \epsilon$$

Intuition: starting from a moment, **every** member in the sequence are close.
Remember, start proof with: given $\epsilon > 0$, choose $N >$ (something related to ϵ, find later), then prove satisfy inequality. But in rough work do opposite.

Theorem 6.7. Cauchy sequences $\Leftrightarrow$ Convergent sequence, they are exactly the same.

Definition 6.8. Given a sequence (a_n), a **series** is defined to be $\sum_{n=1}^{\infty} a_n$, a series convergences when the **partial sum sequence** S_n, defined to be:

$$S_n = a_1 + a_2 + ... + a_n$$

converges. If (and only if) $\lim_{n\to\infty} S_n = L$ we write $\sum_{n=1}^{\infty} a_n = L$

Other than telescoping series (split up partial fractions) or working from definition, the most useful test to conclude series convergence or divergence are as follows:

Theorem 6.9 (Divergence Test)**.** $\lim_{n\to\infty} a_n \neq 0 \implies \sum_{n=1}^{\infty} a_n$ is divergent

For convergence sequence, most of the time we are talking about $a_n \geqslant 0$ for all $n \in \mathbb{N}$, then partial sum sequence S_n is monotone increasing, hence sometimes check bounded is enough.

Theorem 6.10 (Comparison Test)**.** If we want to know a series $\sum_{n=1}^{\infty} a_n$ is divergent or not, then we consider a series (b_n): (Always consider this first, this is the most powerful!)

- $\sum_{n=1}^{\infty} b_n$ convergent and $0 \leqslant a_n \leqslant b_n$ for all $n \in \mathbb{N} \implies \sum_{n=1}^{\infty} a_n$ is convergent.
- $\sum_{n=1}^{\infty} b_n$ divergent and $0 \leqslant b_n \leqslant a_n$ for all $n \in \mathbb{N} \implies \sum_{n=1}^{\infty} a_n$ is divergent.

Theorem 6.11 (k-test)**.** $k > 1 \implies \sum_{n=1}^{\infty} \frac{1}{n^k}$ is convergent.

Theorem 6.12 (Ratio Test)**.** Given a series $\sum_{n=1}^{\infty} a_n$, consider the quantity $A_n = \dfrac{a_{n+1}}{a_n}$, given $A = \lim_{n\to\infty} A_n$, if $A < 1$ then $\sum_{n=1}^{\infty} a_n$ is convergent, $A > 1$ $\sum_{n=1}^{\infty} a_n$ is divergent. Any other case, then the test is inconclusive.

Also, check 'Root Test' in problem set.

7 Functions

Definition 7.1. A value $a \in \mathbb{R}$ is an **accumulation point** of an interval $A \subseteq \mathbb{R}$ iff there exists a convergent sequence (x_n), where $x_n \in A, (x_n) \neq a$ for all n.

Intuitively, that means accumulation point are 'not isolated', in an interval, all points are accumulation point. We proceed with similar definition with sequences. From now on, we will always be talking about $f : A \to \mathbb{R}$

Definition 7.2. A function f is **bounded above** if for all $x \in A$, there exists $M \in \mathbb{R}$ such that $f(x) \leqslant M$, **bounded below** when $f(x) \geqslant M$, again if a function is bounded above **AND** below, then it is bounded, otherwise it is unbounded.

Now we talk about limits to infinity, the definition is very similar to that of sequences.

Definition 7.3. Given A (the set) is unbounded above (for definition purpose)
The **limit to infinity** of a function is defined as the following:

- If for all $\epsilon > 0$, there exists $r, \in \mathbb{R}, x \in A$ such that $x \geqslant r \implies |f(x) - L| \leqslant \epsilon$, then $\lim\limits_{x\to\infty} f(x) = L$
- If for all $M \in \mathbb{R}$, there exists $r \in \mathbb{R}, x \in A$ such that $x \geqslant r \implies f(x) \geqslant M$, then $\lim\limits_{x\to\infty} f(x) = \infty$,
- If for all $M \in \mathbb{R}$, there exists $r \in \mathbb{R}, x \in A$ such that $x \geqslant r \implies f(x) \leqslant M$, then $\lim\limits_{x\to\infty} f(x) = -\infty$

Now the moment is not $n \geqslant N$, but $x \geqslant r$ instead. But general theory is the same.

Theorem 7.4 (Heine's Definition I)**.** A function f with $\lim\limits_{x\to\infty} f(x) = L \Leftrightarrow$ For **all** sequences (X_n) with $\lim\limits_{n\to\infty} X_n = \infty, X_n \in A$ for all n, we have $\lim\limits_{n\to\infty} f(X_n) = L$ (L can be infinity?)

Now we move on to limit at a point: note that the limit exists at $x = a$ is independent with the **value** of $f(a)$ (but it must be defined due to the definition of a function)

Definition 7.5. Given an accumulation point $a \in A$, then the **limit of f at the point** $x = a$ is defined as: Given $\epsilon > 0, r \in \mathbb{R}$, there exists $\delta > 0$ such that for $x \in A$:

- If $0 < |x - a| \leqslant \delta \implies |f(x) - L| \leqslant \epsilon$, then we write $\lim\limits_{x\to a} f(x) = L$
- If $0 < |x - a| \leqslant \delta \implies f(x) \geqslant r$, then we write $\lim\limits_{x\to a} f(x) = \infty$
- If $0 < |x - a| \leqslant \delta \implies f(x) \leqslant r$, then we write $\lim\limits_{x\to a} f(x) = -\infty$
- If $|x - a| \leqslant \delta \implies |f(x) - f(a)| \leqslant \epsilon$, then f is **continuous** at $x = a$

Again, now the moment is δ around a. Note that the larger than 0 argument disappears because we care about the value of $f(a)$ when discussing continuity.

Theorem 7.6 (Heine's Definition II)**.** If a is an accumulation point of A, then Writing $\lim\limits_{x\to a} f(x) = L$ is **equivalent to**: For all sequences (x_n) with $x_n \in A \setminus \{a\}$, $\lim\limits_{n\to\infty} x_n = a$, we have $\lim\limits_{n\to\infty} f(x_n) = L$.

Theorem 7.7 (Heine's Definition III)**.** A function f is continuous at $a \Leftrightarrow$ For all sequences (x_n) with $x_n \in A$, $\lim\limits_{n\to\infty} x_n = a$, we have $\lim\limits_{n\to\infty} f(x_n) = f(a)$.

Definition 7.8. A function $f : A \to \mathbb{R}$ is continuous if it is continuous for all $a \in A$.

Theorem 7.9. Both continuous functions and limits enjoy the "limit laws" in theorem 5.5, namely, addition and multiplication of functions have their expected limits, and they are operations closed in the class of continuous functions.

8 EVT, IVT, More on functions

In the following two theorem, we assume $a, b \in \mathbb{R}, a < b, f(a) < f(b)$.

Theorem 8.1 (Extreme Value Theorem)**.** Given $f : [a, b] \to \mathbb{R}$, then

- f is bounded, and
- There exists $x_1, x_2 \in [a, b]$, where

$$f(x_1) = \sup\{f(c) : c \in [a, b]\}$$
$$f(x_2) = \inf\{f(c) : c \in [a, b]\}$$

Which means the maximum and minimum value of the function are in the function's domain.

Theorem 8.2 (Intermediate Value Theorem)**.** If $f : [a, b] \to \mathbb{R}$, then for all $L \in (f(a), f(b))$ (open interval), there exists $x \in (a, b)$ such that $f(x) = L$.